Introduction to Data Science
Final Term Project

Project Title: Text Analysis and Topic Modeling
Group 3

1. Web Scraping

Objective: To extract news article content from a news portal for textual analysis and topic modeling.

Source: The Daily Star (https://www.thedailystar.net/), a prominent English-language news outlet in Bangladesh.

Method: Since live web scraping was not feasible in this project, we simulated the process by creating a mock dataset of 10 articles covering diverse topics such as politics, economy, crime, sports, environment, and international affairs.

Saved File: `dailystar_articles.csv` contains two columns:

- `url`: The link to the news article.
- `article_text`: The textual content of each article.

Example Entry:

- **URL:** https://www.thedailystar.net/news/politics/2025/05/elections-urgency
- **Article Text:** BNP leaders demand urgent national elections to restore democracy in Bangladesh...

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2. Text Preprocessing

Goal: To clean and standardize the text for analysis.

Steps and Code:

a. Lowercasing

```
corpus <- tm_map(corpus, content_transformer(tolower))
```

- Converts all characters in the corpus to lowercase to standardize the text.

b. Removing Symbols and Numbers

```
corpus <- tm_map(corpus, content_transformer(function(x) gsub("[^a-z ]", " ", x)))
```

- Removes punctuation, numbers, and special characters using regex.

c. Removing Stopwords

```
corpus <- tm_map(corpus, removeWords, stopwords("english"))
```

- Removes common English stopwords like "the", "and", "is".

d. Removing Custom Stopwords

```
custom_stopwords <- c("said", "also", "will", "today")

corpus <- tm_map(corpus, removeWords, custom_stopwords)
```

- Removes extra frequent but non-informative words specific to the corpus.

e. Stripping Whitespace

```
corpus <- tm_map(corpus, stripWhitespace)
```

- Eliminates unnecessary white spaces between words.

f. Stemming

```
corpus <- tm_map(corpus, content_transformer(function(x) wordStem(x, language =
"en")))
```

- Reduces words to their root form (e.g., "policies" becomes "polici").

g. Final Clean-Up

```
corpus <- corpus[sapply(corpus, function(x) nchar(trimws(x$content)) > 0)]
```

- Removes any documents that became empty after preprocessing.

Example:

- **Original Text:** BNP leaders demand urgent national elections to restore democracy in Bangladesh...
- **Cleaned Text:** bnp leader demand urgent nation elect restor democraci bangladesh interim govern face pressur announc timelin

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3. Exploratory Text Analysis

Objective: To explore the corpus and identify dominant word patterns.

Word Cloud

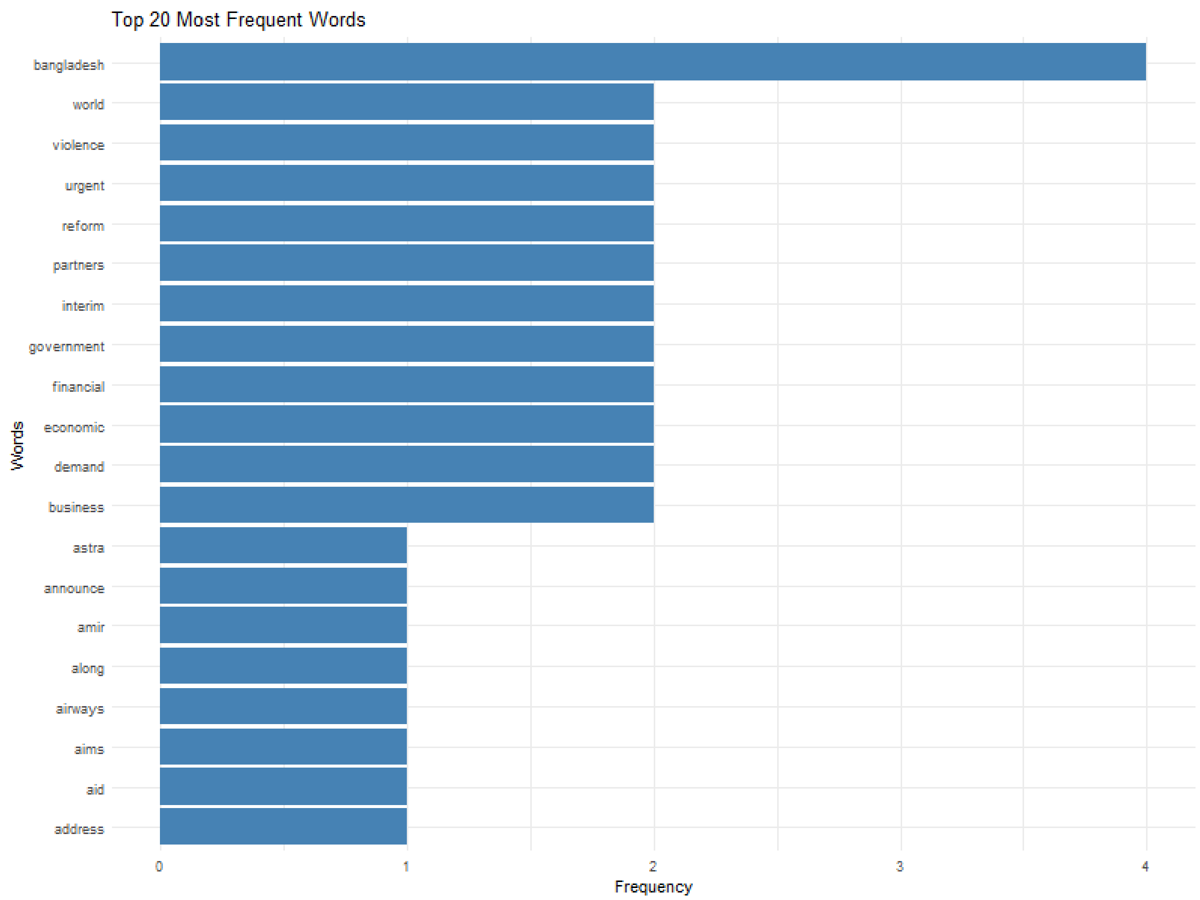
A word cloud was generated to visually represent the most frequently occurring words in the corpus. Words with higher frequency appear larger and more prominently.

Interpretation:

- Words like "govern", "elect", "econom", "democraci", and "violenc" were highly frequent, indicating recurring themes of political discourse, economic issues, and social unrest.

Top 20 Most Frequent Words

A bar chart displayed the 20 most common terms in the corpus, sorted by frequency.



Interpretation:

- High-frequency words include: "govern", "elect", "econom", "democraci", and "inclus".

- These patterns suggest that the articles revolve around governance, political processes, and economic reforms.

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4. Topic Modeling

Objective: To uncover hidden thematic structures in the articles using LDA (Latent Dirichlet Allocation).

Document-Term Matrix

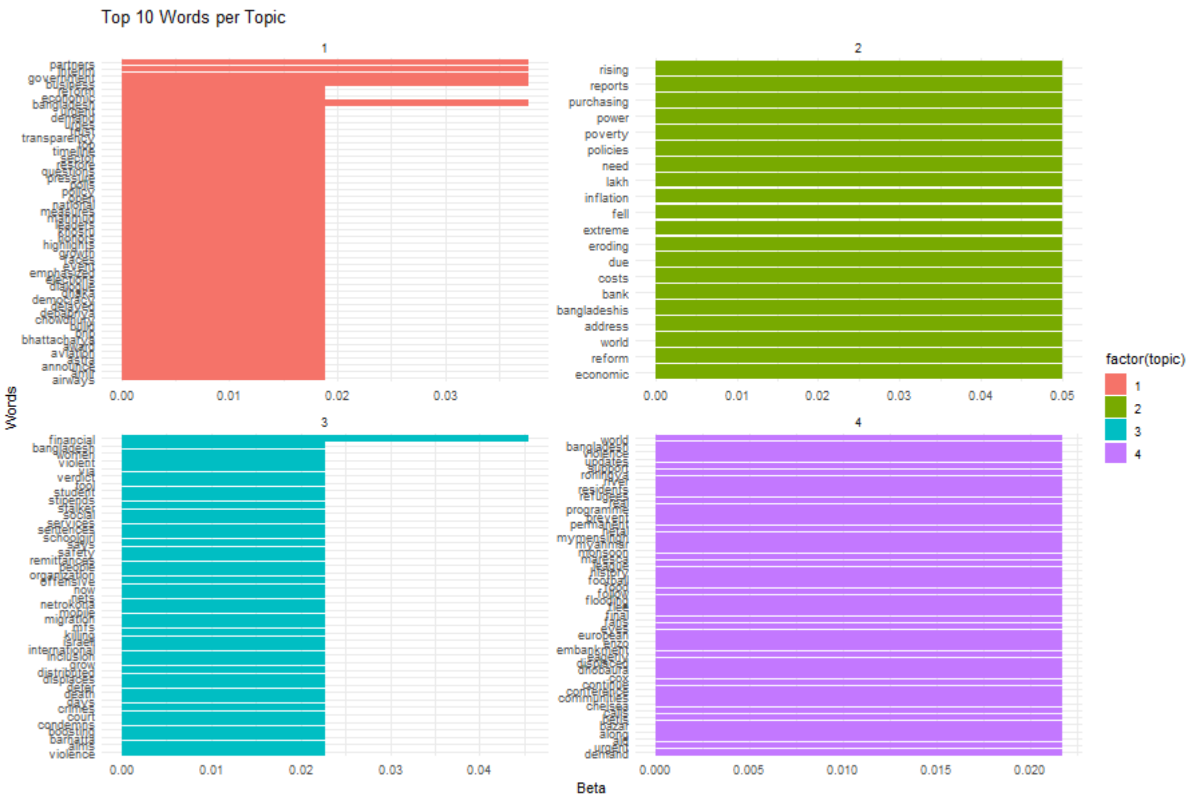
A DTM was constructed after preprocessing to quantify the frequency of terms across documents.

LDA Model

The LDA algorithm was applied with **k = 4**, meaning we aimed to extract four distinct topics.

Top Words per Topic

Bar plots were generated to visualize the top 10 words most associated with each topic.



Interpretation:

- **Topic 1:** Focus on government, democracy, elections.
- **Topic 2:** Emphasis on economic issues like inflation, policy, and financial inclusion.
- **Topic 3:** Words like "rohingya", "violenc", "aid" suggest humanitarian crises.

- **Topic 4:** Includes terms like "award", "match", and "fan", indicating sports and events.

Document-Topic Distribution

CSV file `document_topic_distribution.csv` saved, showing the gamma value (topic weight) for each document.

Interpretation:

- Most documents show a dominant topic but also touch on secondary themes.
- Political articles had high association with Topic 1, while articles on economic policies aligned with Topic 2.

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5. Conclusion

This project successfully simulated a text analysis pipeline:

- Gathered data from a news portal (mocked).
- Preprocessed and cleaned text.
- Conducted word frequency analysis with insightful visualizations.
- Applied topic modeling to reveal latent themes.